

Grade 6 Accelerated
Unit 13
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

Determine if each pair of expressions are equivalent.

1. $-9(20 + 7)$ and $-9(20) - 9(7)$
 $= 9(27)$ $- 180 - 63$
 $= -243$ $- 180 + -63$
 $= -243$
 $-243 = -243$
Equivalent

2. $4.5(10 + 7)$ and $4.5 \cdot 10 + 4.5 \cdot 7$
 $4.5(17)$ $45 + 31.5$
 $= 76.5$ $= 76.5$
 $76.5 = 76.5$
Equivalent

3. $\frac{3}{4} + \frac{9}{10}$ and $\frac{9}{10} + \frac{3}{4}$
 $\frac{30}{40} + \frac{36}{40}$ $\frac{36}{40} + \frac{30}{40}$
 $\frac{66}{40}$ $\frac{66}{40}$
 $= 1\frac{26}{40}$ $= 1\frac{26}{40}$
 $1\frac{13}{20} = 1\frac{13}{20}$
or
 $1.65 = 1.65$
Equivalent

4. $\frac{2}{3} - \frac{1}{2}$ and $\frac{1}{2} - \frac{2}{3}$
 $\frac{4}{6} - \frac{3}{6}$ $\frac{3}{6} - \frac{4}{6}$
 $\frac{1}{6} \neq -\frac{1}{6}$
They are not equivalent.

5. $\frac{1}{6} \times \frac{2}{9}$ and $\frac{2}{9} \times \frac{1}{6}$
 $\frac{1 \cdot 2}{6 \cdot 9}$ $\frac{2 \cdot 1}{9 \cdot 6}$
 $\frac{2}{54} = \frac{2}{54}$
Equivalent

6. $\frac{7}{11} \div \frac{1}{3}$ and $\frac{1}{3} \div \frac{7}{11}$
 $\frac{7}{11} \cdot \frac{3}{1}$ $\frac{1}{3} \cdot \frac{11}{7}$
 $\frac{21}{11}$ $= \frac{11}{21}$
 $= 1\frac{10}{11}$
 $1\frac{10}{11} \neq \frac{11}{21}$
They are not equivalent.

$$\begin{aligned}
 7. \quad & \frac{5}{7} - \left(\frac{1}{2} - \frac{3}{4} \right) \text{ and } \left(\frac{5}{7} - \frac{1}{2} \right) - \frac{3}{4} \\
 & \frac{5}{7} - \left(\frac{2}{4} - \frac{3}{4} \right) \quad \left(\frac{10}{14} - \frac{7}{14} \right) - \frac{3}{4} \\
 & \frac{5}{7} - \left(-\frac{1}{4} \right) \quad \frac{3}{14} - \frac{3}{4} \\
 & \frac{20}{28} + \frac{7}{28} \quad \frac{6}{28} - \frac{21}{28} \\
 & = \frac{27}{28} \quad = -\frac{15}{28}
 \end{aligned}$$

They are not equivalent.

$$\begin{aligned}
 8. \quad & \left(\frac{1}{5} \div \frac{2}{5} \right) \div \frac{8}{9} \text{ and } \frac{1}{5} \div \left(\frac{2}{5} \div \frac{8}{9} \right) \\
 & \frac{1}{5} \cdot \frac{5}{2} \cdot \frac{9}{8} \quad \frac{1}{5} \div \left(\frac{2}{5} \cdot \frac{9}{8} \right) \\
 & \frac{5}{10} \cdot \frac{9}{8} \quad \frac{1}{5} \div \left(\frac{18}{40} \right) \\
 & \frac{45}{80} \quad \frac{1}{5} \cdot \frac{40}{18} \\
 & \quad \quad \frac{40}{90} \\
 & = \frac{9}{16} \quad = \frac{4}{9}
 \end{aligned}$$

They are not equivalent.

9. Rewrite $(y - 11) + 26$ using the associative property.

$$y + (-11 + 26)$$

10. Rewrite $z + 14$ using the commutative property.

$$14 + z$$